Integration, Final Project Results You Can Explain

Bring what you learned into one project



Financial ML · Week 16

This week runs in four steps

From basics to core concepts, cases, then summary

01	**Basics**	**Splits, baselines** Train, validation, test; baselines
02	**Core concepts**	**Five project steps** Tracks, metrics, costs, schedule
03	**Cases**	**Field models, checklist** A model's business failure, final checklist
04	**Summary**	**Honest is good** Check Q&A

Trustworthy results matter more than flashy models

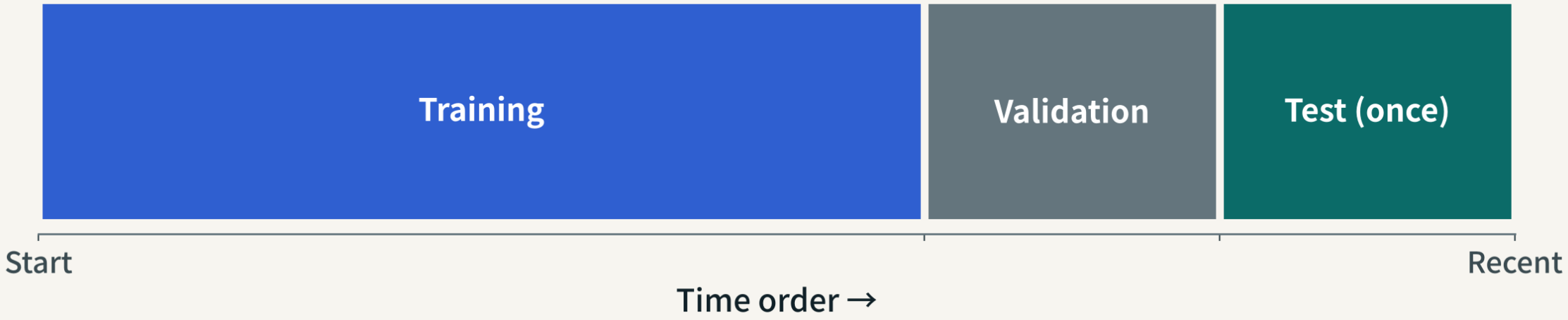
01

Basics

Set splits and baselines before the project starts

Training, validation and test have different roles

With time data, keep this order in time



Tune on validation; view the test once, at the end

Fix after seeing the test, and it is no longer a test

A baseline is the yardstick of model value

Baseline: results from the simplest method

Definition

- Results with no complex model.

Track examples

- Fraud: all normal; investing: equal weight.

Why it matters

- The gain over baseline is the model's real value.

98.6% without a baseline tells you nothing

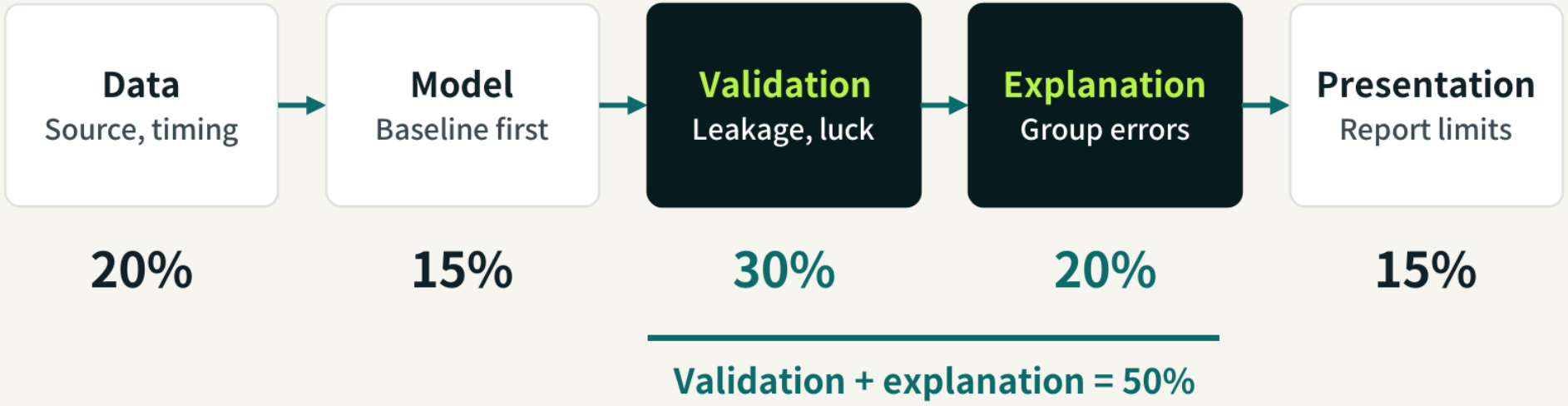
02

Five project steps

Source concepts: grading, tracks, metrics, costs, schedule, talks

The goal is a trustworthy process, not one good number

Repository's sample scoring — question → data → model → validation → explanation → talk



Validation + explanation = 50%

A well-validated simple model is a fine project

Tracks differ in question and evaluation

Pick one small question; start from a simple baseline

Track	Starting question	Baseline	Evaluate
Fraud	**Which trades to review?**	**Logistic reg.**	**Precision, recall, PR-AUC**
Chatbot	**Answer from given docs?**	**Keyword search**	**Retrieval fit, grounding**
Strategy	**Useful after costs?**	**Equal weight**	**Net, out-of-sample**

Time-dependent and i.i.d. classification data need different validation

A model with 98.6% accuracy misses 60% of fraud

20 frauds in 1,000 — 8 caught, 2 false alarms on normal, 12 frauds missed

	Model: fraud	Model: normal
Actual fraud 20	**8** Fraud caught TP	**12** Fraud missed FN
Actual normal 980	**2** False alarm FP	**978** Normal correct TN

Four metrics

$$\text{accuracy} = \frac{8 + 978}{1000} = 98.6\% \qquad \text{precision} = \frac{8}{8 + 2} = 80\%$$

$$\text{recall} = \frac{8}{8 + 12} = 40\% \qquad F1 = \frac{2 \times 8}{2 \times 8 + 2 + 12} = 53.3\%$$

- Calling all normal still gives 98% accuracy.
- Pick metrics by alert capacity and missed-fraud cost.

‘98.6%, so nearly perfect’ is a wrong reading

Read investment results after costs

Edge 30bp, cost 12bp per trade, 40 trades a year — simple sum, same full exposure

1bp = 0.01%p

simple net return approx. $= (30 - 12)\,\text{bp} \times 40 = 720\,\text{bp} = 7.2\%$

Gross edge	Cost	Net edge	Sharpe (10% vol.)
12%	**4.8%**	**7.2%**	**1.2 → 0.72**

※ **Real returns need exposure, overlap, compounding, cost shifts, turnover. Not a general formula**

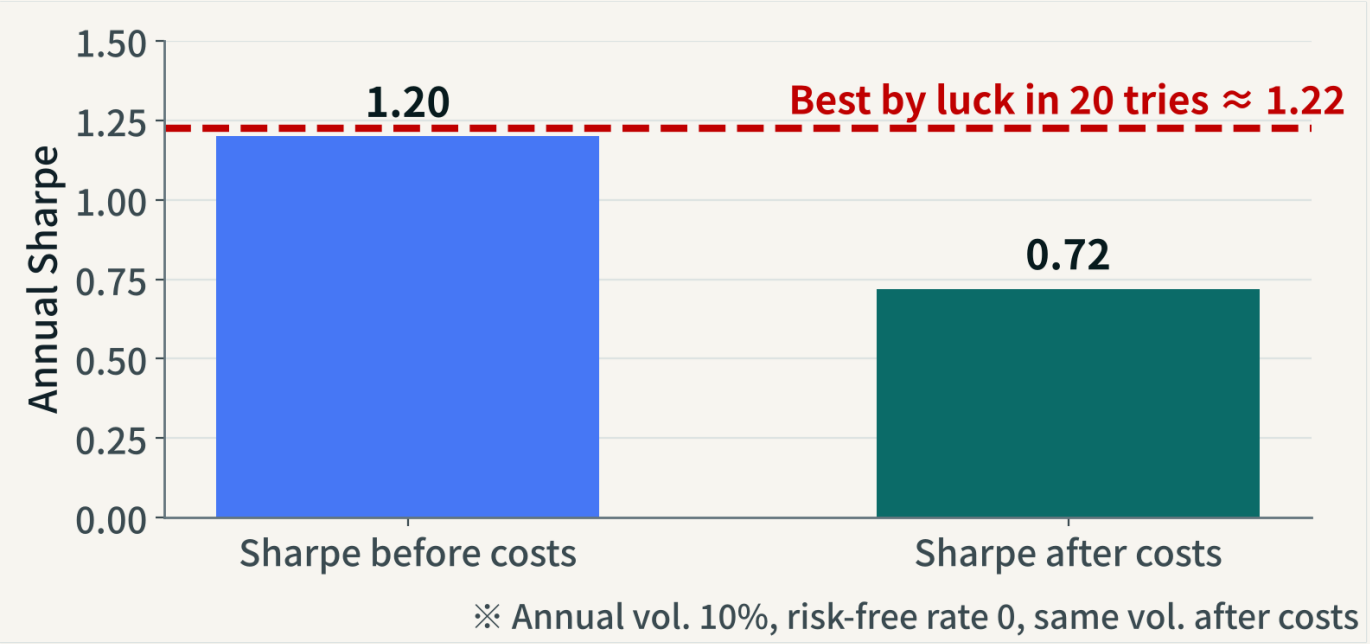
Reporting pre-cost Sharpe alone overstates a strategy

A best Sharpe of 1.2 alone hardly proves skill

20 strategies tried; SD of no-skill Sharpe estimates 0.5

Luck-only max

- Luck-only max of 20 trials is about 1.22.
- Best Sharpe 1.2 is close to it.



Not a formal DSR or pass mark

Disclose trial counts so numbers can be read

A function's name does not ensure time-ordered validation

Source note — StratifiedKFold(shuffle=False), p.31

STRATIFIEDKFOLD

Class balance

- Matches class ratios per fold.
- Does not ensure past-only training.

shuffle=False ≠ time-order validation

Time-dependent

Own design

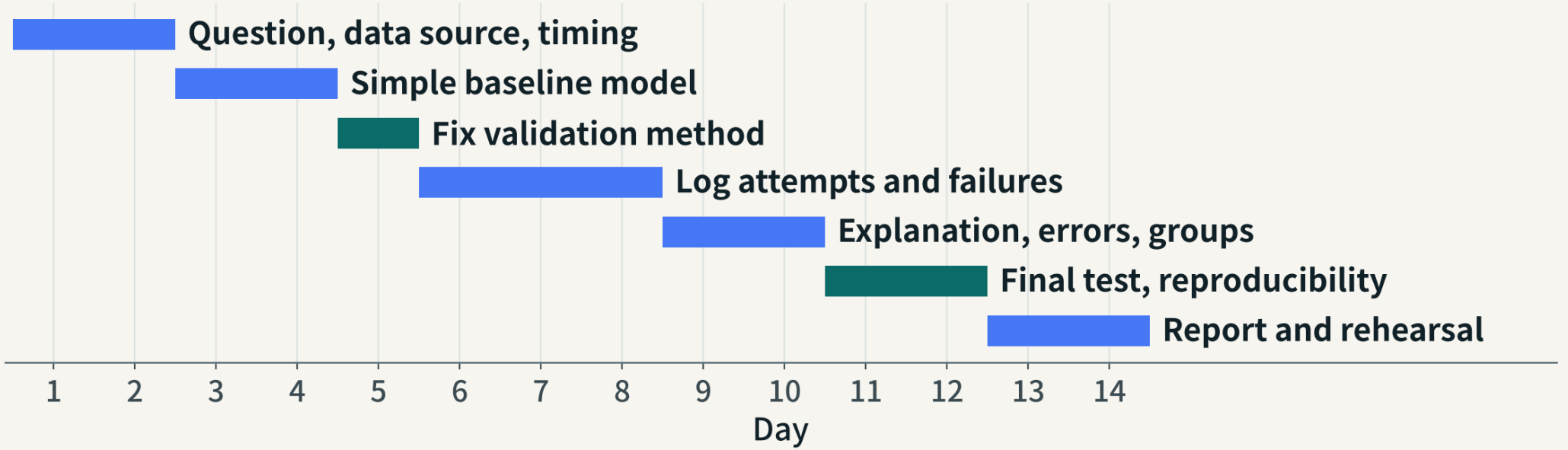
- Design time splits and purging.
- Do not reuse i.i.d. examples.

Fit the data's time structure

Read the implementation, not the algorithm name

A 2-week plan fixes the test and runs it only once

14-day plan for non-majors



※ Dark bars = steps where the test set is fixed and used once

Revised after the final test? Then the new result is not an independent final test

Explaining poor results with evidence is a real achievement

A 10-minute talk gives most time to validation and limits

Question → data, method → validation, results → attribution, errors → limits, next steps



AI use rule

- Use AI, but you must run and explain it yourself.

Failure counts

- Showing why it fails is a meaningful result.

No penalty for not using a complex model

03

Lessons from cases

Lessons from deployed models; pre-submission checklist

A deployed forecasting model ended a business after a market shift

Zillow Offers (2021): buying and selling homes on price forecasts

What happened

- Forecast errors and market shifts cost hundreds of millions; it shut down.
- It announced a ~25% staff cut.

For your project

- Past validation may fail after a regime change.

Lesson

- Report forecast ranges and limits; set risk limits.

Validation continues in operation; it is not the end

Self-check seven items before submitting

The source Unit 16's AI review prompt as a checklist

Item	Question to ask
Question, data	**Are question, data and timing stated?**
Baseline	**Beat a simple baseline?**
Split	**Does the split fit the time structure?**
Leakage	**Any future info, label overlap or test reuse?**
Trials, costs	**Trials and trading costs disclosed?**
Explanation, fairness	**Explanations and group errors read without hype?**
Replication	**Replication and limits written?**

04

Summary

Close the course with key Q&A

An honestly validated result is good

One question, one answer for this week

Check question	Answer
Losing to the baseline = failure?	**Honestly showing what fails is still a meaningful project.**
Can I submit AI-written code?	**Only if you can run, read and explain it yourself.**
Is 98.6% accuracy a good model?	**With class imbalance, check precision, recall and the baseline.**
shuffle=False means time-ordered?	**No. Design the time split and purging separately.**

Remember: validation and explanation are half the grade

MODELS ARE CHEAP.

JUDGEMENT IS NOT.

VALIDATE, THEN EXPLAIN.

Validate honestly,
claim only results you can explain.